MTH357

Timed Online Assignment
(January 2021 Semester- Answer Keys):
For students' practice and reference

Question 1 (Total: 34 marks)

The quality of red wine is thought to be related to the properties of clarity, aroma, body, flavour and oakiness. 38 red wines were evaluated by experts and data summary is shown in Table Q1-I and Figure Q1 below:

Index No	Clarity	Aroma	Body	Flavour	Oakiness	Quality	Region
1	0.5	3.3	5.4	4.3	3.6	12.3	2
2	0.7	7.7	6.6	6.7	3.7	16.1	3
3	0.8	5.9	5.7	7	4.1	16.1	3
…	…	…	…	…	…	…	…
38	0.9	3.4	5	3.4	3.4	7.9	2

Table Q1 - I

```
   Clarity           Aroma            Body          Flavour
Min.   :0.5000   Min.   :3.300   Min.   :2.600   Min.   :2.900
1st Qu.:0.8250   1st Qu.:4.125   1st Qu.:4.150   1st Qu.:4.225
Median :1.0000   Median :4.650   Median :4.750   Median :4.800
Mean   :0.9237   Mean   :4.847   Mean   :4.684   Mean   :4.768
3rd Qu.:1.0000   3rd Qu.:5.450   3rd Qu.:5.375   3rd Qu.:5.500
Max.   :1.0000   Max.   :7.700   Max.   :6.600   Max.   :7.000
   Oakiness        Quality          Region
Min.   :2.900   Min.   : 7.90   Min.   :1.000
1st Qu.:3.700   1st Qu.:11.15   1st Qu.:1.000
Median :4.100   Median :12.45   Median :2.000
Mean   :4.255   Mean   :12.44   Mean   :1.868
3rd Qu.:4.775   3rd Qu.:13.75   3rd Qu.:3.000
Max.   :6.000   Max.   :16.10   Max.   :3.000
```

Figure Q1

The summary of the regression results from R is shown below in Table Q1-II:

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	4.00	2.23		0.08
Clarity	2.34	1.73		0.19
Aroma	0.48	0.27		0.09
Body	0.27	0.33		0.42
Flavour	1.17	0.30		0.00
Oakiness	-0.68	0.27		0.02

Table Q1 - II

R^2: 0.7206
$R^2 - adj$: 0.6769
F: 16.51 on 5,32 $DF, p - value <$ 0.0001

(a) The data is stored in the folder "D:/SUSS2021/MTH357" under "Wine Data.csv". Write down the R code which will read the data from the file, generate the summary statistics and fit the multiple linear regression model above. Note that the model above does not include "Region" as an independent variable.

(8 marks)

(b) Write down the fitted regression model. Interpret the coefficients for Flavour and Oakiness, and discuss the limitations to the model.

(7 marks)

(c) Compute the missing t-values in Table Q1-II, and perform t-tests to assess the contribution of each regressor to the model. What conclusions can you draw?

(8 marks)

(d) The analyst decided to exclude the variable "Body" from the model. (i.e. the new model has Clarity, Aroma, Flavour and Oakiness as X variables). What will happen to the R^2 and $adj\ R^2$? Write down a possible value for the new R^2 and $adj\ R^2$, and explain your answer clearly.

(8 marks)

(e) There are a number of variables which are not statistically significant under the individual t-tests. If we were to use nested model F-test instead to evaluate if the variable is worth including in the model, will we always reach the same conclusion as the t-test?

(3 marks)

Marking Guide:

(a) The data is stored in the file "Wine Data.csv".

```
setwd("D:/SUSS2021/MTH357")
data <- read.csv("Wine Data.csv",header=TRUE")
summary(data)
```

Steps above are relatively straightforward without good alternatives.

```
data <- data[,!names(data) == "Region"]
mod1 <- lm(Quality ~ . , data=data)
summary(mod1)
```

(b) The student needs to express the points Ys = f(X) + error.

$$Quality_i = 4 + 2.34 * Clarity_i + 0.48 * Aroma_i + 0.27 * Body_i + 1.17 * Flavour_i - 0.68 * Oakiness_i + \varepsilon_i$$

Y-hat = f(Xi)

$$\widehat{Quality}_i = 4 + 2.34 * Clarity_i + 0.48 * Aroma_i + 0.27 * Body_i + 1.17 * Flavour_i - 0.68 * Oakiness_i$$

Coefficient Flavour: The expected quality is 1.17 points higher for every increment in flavour points. As flavour has a min and max of 2 and 7 points respectively, the impact of flavour cannot exceed this range.

Coefficient Oakiness: The expected quality is 0.68 points lower for every increment in Oakiness points. As Oakiness has a min and max of 2 and 6 points respectively, the impact of Oakiness cannot exceed this range.

(c)

	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	4.00	2.23	**1.794**	0.08
Clarity	2.34	1.73	**1.353**	0.19
Aroma	0.48	0.27	**1.778**	0.09
Body	0.27	0.33	**0.818**	0.42
Flavour	1.17	0.30	**3.9**	0.00
Oakiness	-0.68	0.27	**-2.519**	0.02

The same t-test is performed for all 6 coefficients.
Ho: The coefficient is equal to 0.
H1: The coefficient is not equal to 0.

We can set alpha = 5%. (or any other acceptable value)

t-statistics are calculated above. And the p-values, calculated from t-distribution with 32 df are provided. Coefficients with low p-values are statistically significant. This means only intercept, flavour and oakiness are significant (Ho is rejected).

(d) R^2 always increase even if the new variable is not statistically significant. However, the increment is very little since the model already has a few good features. We can expect the current R^2 to be very slightly lowered. From the original 0.72, we expect it to drop very slightly to around 0.715. (Decrease of 0.5%)

$Adj\ R^2$ can decrease if the new variable is not statistically significant. It comprises of 2 factors – an increase due to increased R^2, and a decrease due to a penalty on number of variables. Since we expect R^2 to barely move, $Adj\ R^2$ will similarly move very little, probably even less than R^2's movement. From the original 0.677, we can expect it to move up very little, to 0.678 – 0.682 or so. (Increase of 0.1% to 0.5%, cannot be more than R^2's movement)

(e) The F-test and t-test are only equal when we consider simple linear regression. When we move into multiple linear regression, the results can differ, albeit not frequent, especially when we consider multiple coefficients at once.

Question 2 (Total: 16 marks)

(a) Explain Cook's Distance and how can it be used to assess the model in regression analysis.

(4 marks)

(b) The Cook's Distance from the Regression in Question 1 was computed and visualized in the Figure Q2(b) below. The dependent variable is Quality, and while the independent variables are clarity, aroma, body, flavour and oakiness. Find the appropriate F value and hence evaluate the Cook's Distance.

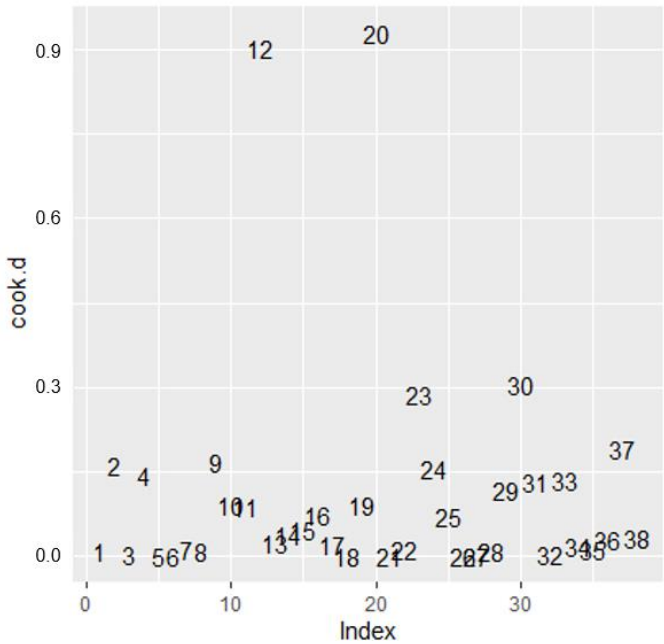


Figure Q2(b)

(5 marks)

(c) One of your colleagues felt that the points 12 and 20 in Figure Q2(b) above should be excluded from the analysis. Is it best practice to exclude data which might be outliers? Discuss.

(3 marks)

(d) In simple linear regression, the standard error of the OLS estimate for slope is given by:

$$\sigma^2_{\hat{\beta}_1} = \frac{\sigma^2}{S_{XX}}$$

where σ^2 may be estimated by

$$s^2 = \frac{SSE}{n-2}.$$

By considering both numerator and denominator, discuss what happens if the sample size is increased by a large amount (towards infinity). Can the standard error of the slope be reduced infinitesimally close to 0 with a greater sample size?

(4 marks)

<u>Marking Guide:</u>

(a) Cook's distance is used to estimate the influence of a data point when performing regression analysis. Cook's distance is computed by removing a single row of data, and by comparing the squared difference of the new predicted y versus the predicted y if that row of data is included. And thus, we call this a "deletion diagnostic". Since Cook's distance compares predicted ys, we will get a single Cook's distance metric for every row of data, similar to DFFITS.

(b) F(p+1,n-p-1)=F(5+1,38-5-1)
R: qf(0.5,6,32)=0.91 (usually quite close to 1)

According to Cook's distance, points 12 and 20 are borderline influential.

(c) There are a few considerations before deleting outliers. Firstly, we need to verify if the outliers are true results that we should factor in regardless. For example, the NASA example mentioned in class had extreme values of ozone which are actually true measurements. We can always resort to using more robust models that are less affected by extreme values. Secondly, we should check for the reasons that could result in the outlier. Is it a data collection / entry problem?

(d) As $n \to \infty$,

$$s^2 = \frac{\text{SSE}}{\text{n} - 2}$$

s^2 is the mean square error and it does not reduce to 0 as $n \to \infty$. SSE, the sum of square errors increases as n increases, n-2 also increases as n increases.

S_{XX} is the sum of squares of x-mean(x), which approaches infinity as sample size increases.

Thus, $n \to \infty$, the numerator of $\sigma^2_{\hat{\beta}_1}$ is still a finite number, while $S_{XX} \to \infty$, and thus $\sigma^2_{\hat{\beta}_1} \to 0$. Thus, the standard error of the slope can be reduced infinitesimally close to 0 with a sample size that approaches infinity.

Question 3 (Total: 25 marks)

The Wine Appellation system is a method where countries categorize their wines into specific grape-growing **geographical regions.** If the wine production meet certain stringent conditions, the wines can be associated with an appellation - a stamp of "high quality". For instance, since 1937, France qualify their wines with the Appellation d'Origine Contrôlée (AOC) system. Famous appellations in France include the Champagne & Bordeaux, amongst hundreds of others.

We would like to know if Wine Quality is affected by Region. In the data from Question 1, the wines come from 3 regions, tagged as Region 1, Region 2 and Region 3. Region 3 are the geographical regions with appellations.

(a) Using Regression Analysis, demonstrate how we can investigate if Quality is affected by Region.

For simplicity, assume that we only have Quality (y), Flavour (x_1) and Region (x_2) in our dataset. Flavour is known to affect Quality and must be included in the Regression Model.

Your answer must include (marks in parenthesis):

(i) The Regression Model that you propose to use.

(4 marks)

(ii) The hypotheses which are to be validated.

(3 marks)

(iii) Even without running the regression, you should already have an expectation of how the coefficients should be. Write down the expected signs (positive or negative) and magnitude of the coefficients.

(3 marks)

(iv) The statistical tests to be conducted, stating the distribution clearly.

(4 marks)

(v) How the results may be interpreted?

(2 marks)

Marks will be given for detailed but concise answers. Make sure to explain your reasoning clearly.

(b) Explain the need to include Flavour into the model if we know for a fact that it affects Quality.

(4 marks)

(c) Using the same wine quality data, we fit a model that relates wine Quality (y) to Flavour (x_1) and Region (x_2), but this time, we use Region as an allocated code, taking on the values shown in the data (1, 2 and 3s), instead of treating it as a categorical variable. Discuss the interpretation of the model in this model. Under what situation will this be a reasonable approach?

(5 marks)

<u>Marking Guide:</u>

(a) (i) There are 3 Regions. We will need to recode the Region data into categorical variables. We set Region 3 as the base because it is the area with appellations, i.e. expected highest quality. Thus Region 3 will likely have greatest difference compared to the rest.

$$y_i = \beta_0 + \beta_1 Region1 + \beta_2 Region2 + \beta_3(Flavour - \min(Flavour)) + \varepsilon_i$$

Where Region1 = 1 if Region=1, Region1 = 0 otherwise.
Where Region2 = 1 if Region=2, Region2 = 0 otherwise.

Correct model: 2 marks. Correct recoding of Region: 2 marks.

(ii) $H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$
$H_0: \beta_2 = 0$ vs $H_1: \beta_2 \neq 0$
$H_0: \beta_1 = \beta_2$ vs $H_1: \beta_1 \neq \beta_2$

(iii) We expect β_1 & β_2 to be negative because Region 3 is the baseline in this case. Since Region 3 has wines with appellations, we expect their quality rating to be highest. Quality (y) has values from 7 to 16, and thus, the coefficients cannot be much larger than 8. (8 Means region 3 wines are 8 quality higher than other region, which is almost the max – min of quality).

We have little information of the differences between Region 1 and 2, so it is difficult to tell their relative magnitude.

(obviously, this answer depends on their coding in part i)

(iv) From the regression outputs, we can check the p-values of the β_1 and β_2. These p-values are calculated based on t-34 distribution. (df: 38-3-1). We will reject the null hypothesis if the p-value is lower than 5%.

The easier approach to check for $H_0: \beta_1 = \beta_2$ vs $H_1: \beta_1 \neq \beta_2$ is to form confidence intervals for both β_1 & β_2 and check if they overlap. The confidence intervals can be formed by $\hat{\beta} \pm t_{34,0.975} * SE$.

(v) If $\beta_1 = 0$, we can conclude that there is a statistically significant difference between Region 1 and 3.

If $\beta_2 = 0$, we can conclude that there is a statistically significant difference between Region 2 and 3.

If $\beta_1 = \beta_2$, we can conclude that there is no statistically significant difference between Region 1 and 2.

(b) Including flavour into the regression function will act as an adjustment for the effects of flavour on quality. It removes effects of flavour on quality. $Flavour - \min(Flavour)$ is a number that starts from 0, and thus the coefficient represents the increasing effect of Flavour from baseline.

(c) $y_i = \beta_0 + \beta_1 Region + \beta_3(Flavour - \min(Flavour)) + \varepsilon_i$

Using a model like this, it means that the difference in Quality for Region 3 vs Region 2 is the same as Region 2 vs Region 1, and the difference for Region 3 vs Region 1, is exactly twice of Region 2 vs Region 1. This is very likely a strong assumption.

It is not impossible to use such a model, but it requires a lot of knowledge beforehand on how Regions affect wine quality. For example, Region needs to be at least an ordinal variable, so that we know Quality of Region 3 > Region 2 > Region 1. Then this model makes the additional assumption that the difference between regions are the same.

Question 4 (Total: 25 marks)

In a study conducted in 2017-2018, a research team from the Lee Kuan Yew School of Public Policy estimated that Singapore seniors each need at least $1,379 a month to meet basic needs. This can be fulfilled by a combination of savings and investment.

To estimate if the population is saving sufficiently to meet retirement needs, a researcher interviewed 15 working people and asked for their savings. He wants to analyse the relationship between age (x) and savings (y) by using regression analysis.

$y_i = f(x_i) + \varepsilon_i$, $\varepsilon_i \sim N(0, \sigma^2)$

Here are his findings as seen in Figure Q4 below:

	Age	Savings / month (SGD)
1	46.6	1012
2	43	991
3	35.4	995
4	31.6	958
5	29.3	864
6	28.7	865
7	28	964
8	27.7	852
9	25.1	801
10	23.8	658
11	23.8	700
12	23.7	721
13	23.7	747
14	21.3	653
15	18.9	722

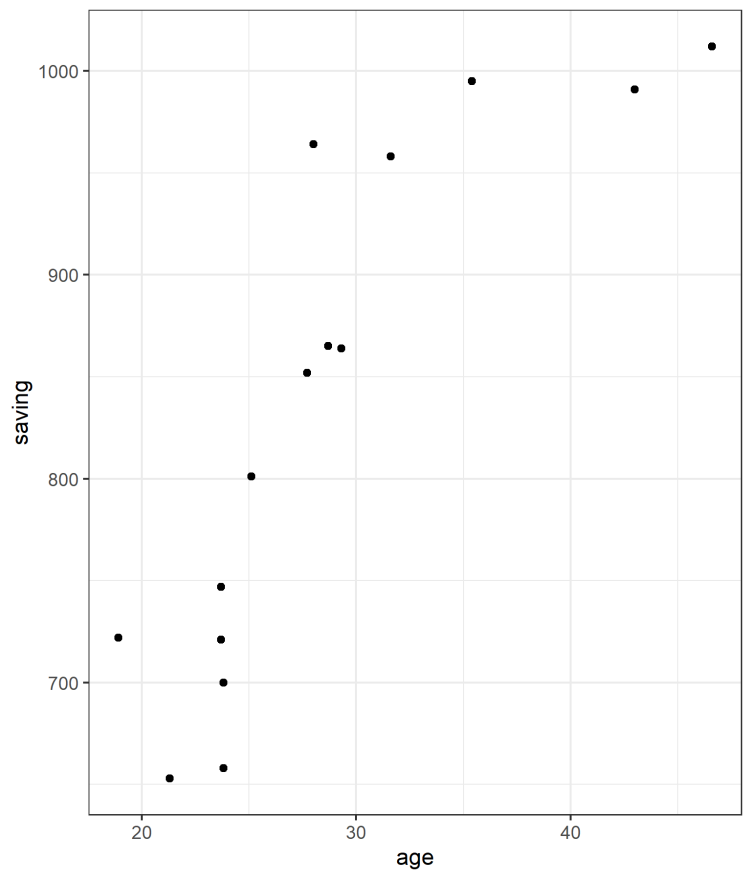


Figure Q4

(a) Calculate the summary statistics and hence compute the coefficients for a simple linear regression model.

(6 marks)

(b) Given that $s = 70.97$ on 13 df, perform a statistical test to verify if β_1 is statistically significant. What assumptions have you made in the statistical test?

(5 marks)

(c) Using the linear regression model obtained in Question 4(a), analyse the difference in the expected savings per month of a 45 year old versus a 25 year old? Construct a 95% confidence interval for the expected difference.

(6 marks)

(d) Using the chart given in the Question or otherwise, draw the (OLS) simple regression line. Comment on the fit.

(4 marks)

(e) The research supervisor suggested to use polynomial terms to improve the model fit. Discuss how can we implement this, the sign of the coefficient (positive or negative) and how can we validate if the new model is superior to the current model?

(4 marks)

<u>Marking Guide:</u>

(a) Sum(Age) = 430.6
Sum(Savings) = 12503
Sum(Age^2) = 13202.92
Sum(Savings^2) = 10654043
Sum(Age * Savings) = 370772.9
Mean(Age) = 28.70667
Mean(Savings) = 833.5333

$$S_{xy} = 370772.9 - \frac{430.6 \times 12503}{15} = 11853.45$$
$$S_{xx} = 13202.92 - \frac{430.6 \times 430.6}{15} = 841.8293$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{11853.45}{841.8293} = 14.08059$$
$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 429.33$$

(b) $\sigma^2_{\hat{\beta}_1} = \frac{\sigma^2}{S_{XX}}$, $s^2 = \frac{SSE}{n-2}$

$$\sigma^2_{\hat{\beta}_1} = \frac{70.97^2}{841.8293}, \sigma_{\hat{\beta}_1} = 2.446036$$

$H_0: \beta_1 = 0$ vs $H_1: \beta_1 \neq 0$
With test statistic: $t_{13} = \frac{14.08059}{2.446036} = 5.757. P - value < 0.0001$.

We can conclude that β_1 is statistically significant. We have made the assumption that $\hat{\beta}_1$ is normal.

(c) $\hat{y} = 429.33 + 14.08059 \times Age$

$$For\ the\ age\ 45, \widehat{y_{45}} = 429.33 + 14.08059 \times 45 = 1062.957$$
$$For\ the\ age\ 45, \widehat{y_{25}} = 429.33 + 14.08059 \times 25 = 781.3447$$

The expected difference is \$281.6123 .

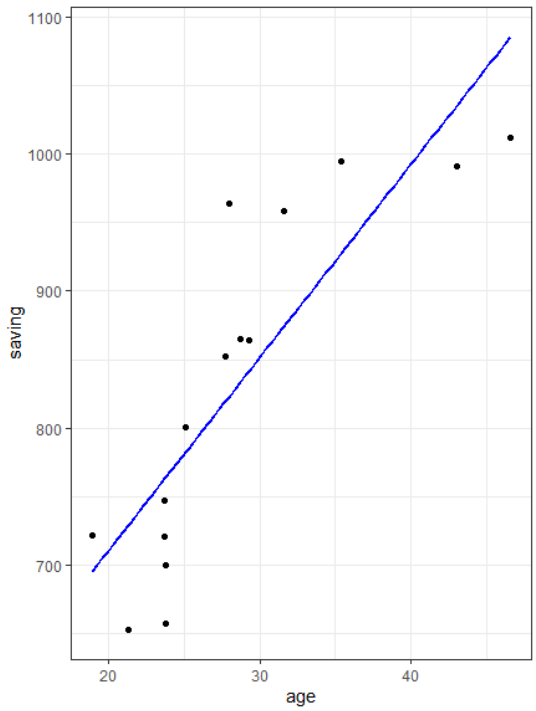
Var($\widehat{y_{45}} - \widehat{y_{25}}$) $= Var(\hat{\beta}_0 + \hat{\beta}_1 \times 45 - \hat{\beta}_0 - \hat{\beta}_1 \times 25) = Var(20\hat{\beta}_1) = 400Var(\hat{\beta}_1) = 400 * 5.983092 = 2392.237$

The standard dev is = 48.92072 .

Qt(0.975,13) = 2.160369

Therefore, the 95% CI is $281.6123 \pm 2.160369 * 48.92072$ or (175.9255, 387.2991).

(d)



While this fit will obtain a good R^2, it is clearly not a good fit because the savings in the middle age group is higher than predicted, while the younger and older age group is under predicted.

(e) While adding in polynomial terms is not difficult, we need to be careful that we do not do that too much that the model overfits. Given that our dataset only has 15 points, I recommend that we keep to Quadratic or Cubic equations.

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \times Age + \hat{\beta}_2 \times Age^2$$

The new coefficient $\hat{\beta}_2$ will be negative, $\hat{\beta}_1$ will be positive, $\hat{\beta}_0$ is unknown, but will surely be smaller than the current value.

We can validate if the new model using goodness-of-fit test, for instance, the nested model F-test, or we can just simply use t-test to validate if the coefficient is statistically significant.

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